

MOSFET

1. $\frac{1}{C_{ox}} + \frac{1}{C_{dep}} = \frac{1}{C}$ (b) $C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$

2. $I_{DS} = \frac{1}{2} \mu \frac{W}{L} C_{ox} [(V_G - V_t) V_{DS} - \frac{V_{DS}^2}{2}]$

3. $I_{DS} (CLM) = \frac{\beta}{2} C_{ox} (V_G - V_t)^2 (1 + \lambda V_{DS})$

4. Subthreshold current $\rightarrow$ enhanced on-current.

$I_{sth} = I_s \exp(\frac{V_G - V_t}{n \cdot V_{thermal}})$

5. On resistance: $R_{on} = \frac{1}{\beta \cdot (V_G - V_t)}$

4.(b) $V_{thermal} = \frac{KT}{q} \ln(10) (60mV)$

6. Velocity saturation

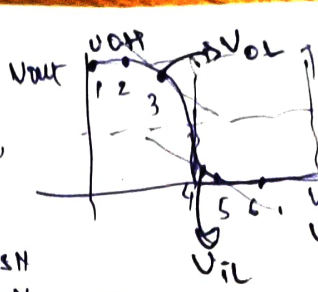
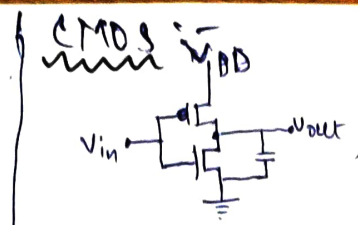
High field charge carriers low V_{DS} dependence. $[v_d = \frac{\mu E}{1 + \frac{\mu E}{v_{sat}}}]$ $E \ll E_c$

($E = \frac{V_{ds}}{L}$) (early saturation)

7. Hot injection & Gate induced drain leakage. (High E field) High k low doped drain.

8. DIBL (drain induced barrier leakage) (SOI, GAA, FinFET).

9. Punch thru effect

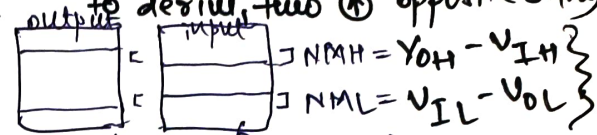


1. NMOS off, PMOS linear.
2. NMOS sat, PMOS linear.
3. NMOS sat, PMOS sat (unstable).
4. NMOS linear, PMOS sat.
5. NMOS linear, PMOS off.

- (i) $I_{DST} = -I_{DSH}$
- (ii) $V_{GSP} = V_{in} - V_{DD}$
- (iii) $V_{DSP} = V_{out} - V_{DD}$
- (iv) $V_{GSN} = V_{in}$
- (v) $V_{DSN} = V_{out}$

(I) $V_M =$ when NMOS sat + PMOS sat $\rightarrow$ (use I_{sat} eqn to deriv. two opposite sides)

(II) Noise margin: (Negative gain)



$\rightarrow$ i) $V_{OH} = V_{DD}$, $V_{OL} = 0$, $V_{ILA} = (V_{out} = -1)$ condition

(III) Skewing: $V_M = \sqrt{\frac{\beta_P}{\beta_N}} (V_{DD} - V_{TP}) + \sqrt{\frac{\beta_N}{\beta_P}} (V_{TN})$

(IV) Power

- i. Dynamic: $\propto C_L V_{DD}^2 f$
- ii. Static: $V_{DD} \cdot (I_{ST} + I_{GOL})$
- iii. Short circuit: $V_{DD} \cdot I_{sc}$ (peak-to-peak)

Photolitho graphy: ArF 193 nm laser, litho-etch-litho-etch (LELE), LFL, SADT (self aligned double patterning).

$\rightarrow$ vapour, adhesion (HMDS), spin coat, bake, (mask, irradiate, bake), (develop, hard bake), etch.

$\rightarrow R = k_1 \frac{\lambda}{NA}$ (R needs to be decreased) (blue light preferred) λ fixed.

$\rightarrow$ Etch (wet: Acid, HF) (Dry: Reactive ion, chemical, plasma). (just ion)

(NH₄OH, H₂O₂, deionized H₂O (water))

full fab process: 1. Clean: organic, inorganic (just add HCl), oxide (HF treatment).

2. Doping: ion implant (ion beam, sputter acc, less magnetic quads), ele scanning. Diffusion (high temp process, D_n d.c., isotropic, low precision, deeper, less control).

3. Deposition: PVD: sputtering / e beam (bombard chemical source, nucleate on substrate). (containing) CVD: high temp, process, Reaction kinetics & fast needed, faster, vacuum reqd, ALD: costlier, slower, precursors sequential, self limiting rxn, deposit-purge chain, (slow).

4. Interconnect: 1. Skin depth $\rightarrow$ eddy current due to AC. unwanted loss & dissipation. MIT: thicker, wider conductors, reduce R .

2. RC delay: (i) low dielectric b/w wires. thicker wires help to reduce R . ($RC = \tau$)

3. Electromigration: caused by current density Δ line, creates voids & hillocks. MIT: Block diffusion / barrier layer: Ta/TaN, Bouffon mechanically (SiN, CoWP). use w vias.

4. Stress migration: due to heat expansion. use annealing to fix.

FEOL $\rightarrow$ transistor making BEOL $\rightarrow$ interconnect skeleton making

FinFET $\rightarrow$ good. improved electrostatic control, reduced leak current, overdrive current higher $\rightarrow$ BAD $\rightarrow$ design complexity, variability, thermal management, higher parasitic C.

SOI: improved subthreshold leakage, BOX (buried oxide) b/w substrate & ft.

FinFETs: Gate is wrapped around channel, increases surface area $\downarrow R$ (Resistance) High k dielectric (HfO₂, ZrO₂) Challenges: self heating (try to confine in plane)

